ROUGH DRAFT

Beneath the Mesa

Assessing Aquifer and Well Water Resources Across Sandoval County, New Mexico

A County-Scale Screening Study with Cuba, New Mexico as a Focus Area

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ABSTRACT

Groundwater supplies most of New Mexico's drinking water, yet long-term monitoring is uneven, especially in rural areas. We assembled eight public datasets from the U.S. Geological Survey, the New Mexico Bureau of Geology and Mineral Resources, and the Office of the State Engineer into one reproducible workflow and used them to screen groundwater conditions across Sandoval County, with the Village of Cuba as a case study. At the county's nine long-term monitoring wells (1998–2026), the water table fell at five, most steeply at Rio Rancho (0.36–0.40 ft per year) and Bernalillo (0.52 ft per year), and was still falling at four of seven wells in 2016–2025; at the Sandia site it reversed and rose about 13 ft after 2007, around the time the Albuquerque area began replacing part of its groundwater pumping with river water. Across 180 wells, deeper water goes with deeper wells (Spearman $\rho \approx 0.63$), and median depth to water ranges from 22 ft near Cochiti to 375 ft near Bernalillo–Rio Rancho; aquifer-test transmissivity has a county median of 136 ft^2/day but varies by almost five orders of magnitude within a single formation. For Cuba, we reconstructed a historical baseline from 15 wells tested in 1957–1979, including water 18–24 ft below land in the Ojo Alamo Sandstone beside the village, and found the one San Jose Formation test consistent with published values. The nearest repeatedly measured well is 38 km from Cuba, so these records cannot yet show a trend there, but they identify which wells and records would. All results are published in an open dashboard and repository.

1 Introduction

1.1 Background and Motivation

Groundwater supplies 78% of New Mexico's drinking water, and 81% of the state's public water systems depend on it (Cook, 2026). The state expects about 25% less water to be available within 50 years (Office of the Governor of New Mexico, 2024), which will increase reliance on aquifers whose condition is often poorly known: rural areas in particular have little groundwater monitoring (Cook, 2026). Sandoval County, home to 148,834 people in 2020 across 3,710 square miles (U.S. Census Bureau, 2026), depends on groundwater for its drinking water: in 2010, public water systems in the county drew about 99% of their 15,900 acre-feet from wells (New Mexico Office of the State Engineer, 2013).

The Village of Cuba (population 630 in 2020; U.S. Census Bureau, 2021), in the county's rural northwest, shows what is at stake. The village supplies its residents from its own wells, which produce about 500,000 gallons (Hollinshead, 2021), and its water system has faced repeated problems: a state emergency declaration after damage to its water infrastructure in 2022 (Office of the Governor of New Mexico, 2022), mineral-rich, sulfurous water that fouls pipes and storage tanks (Davis, 2020; Hollinshead, 2021), a federal partnership to replace its water storage tank (U.S. Army Corps of Engineers, Albuquerque District, 2025), and well-pump outages that forced extreme conservation in the summer of 2026 (Village of Cuba, 2026a,b). This history motivated the concern, stated in our project proposal, that Cuba's principal aquifer may be approaching severe depletion. Failing equipment and poor water quality are not the same as a falling water table, however, and the public record had not been examined for evidence of the latter.

This project tests what public data can show about groundwater in Sandoval County and around Cuba, rather than assuming how severe any decline is or what causes it. It also follows the course's emphasis on peace engineering, "the application of engineering, science, and technology principles to promote and support peace" (University of New Mexico, 2026): an open, reproducible analysis that states its own uncertainty gives a small community and the agencies that serve it a shared factual basis for decisions about a scarce resource.

1.2 Approach

The study works at two levels:

- **County-wide screening.** Map and compare aquifers, wells, and groundwater levels across Sandoval County. Because the county crosses several geologic regions, comparisons are grouped by area and, where the records name it, by aquifer unit, instead of treating the county as one uniform aquifer.
- **Cuba case study.** A closer look at the Ojo Alamo Sandstone and San Jose Formation near Cuba, using published maps of both formations, aquifer-test records from nearby wells, and stream measurements on the Rio Puerco at Cuba.

1.3 Research Questions

RQ1. How have groundwater levels changed at monitored wells in Sandoval County, and which wells show sustained decline, stability, or recovery?

RQ2. What do well records show about well depth, depth to water, and the aquifer units that wells in different parts of the county draw from?

RQ3. What do available records show about groundwater near Cuba, and are they enough to evaluate the concern that its main aquifer is approaching severe depletion?

RQ4. How do the Ojo Alamo Sandstone and San Jose Formation compare near Cuba in extent, thickness, and hydraulic properties?

RQ5. Where is monitoring coverage strong, and where are the most important data gaps?

RQ6. Which indicators can be combined into a dashboard that shows groundwater stress without overstating how certain the data are?

The project proposal also asked about groundwater quality, acequias, and links between groundwater and surface water. Those questions were dropped because the data needed to answer them were not available (Section 7).

1.4 Contributions

Working only with public records, this study:

(1) combined eight datasets into one documented, reproducible workflow, from raw downloads through cleaning to analysis, an interactive dashboard, and this report;
(2) measured the long-term and recent water-level trend at every long-term monitoring well in the county's records;
(3) characterized depth to water at 180 wells and aquifer properties at 108 tested wells, by area and by geologic formation;
(4) reconstructed a historical groundwater baseline for the Cuba area and checked it against published studies; and
(5) mapped where and when monitoring exists, turning the gaps into specific recommendations.

2 Study Area

2.1 Sandoval County

Sandoval County spans several hydrogeologic settings (Fig. 1). Its largest communities, Rio Rancho, Bernalillo, and Placitas, lie in the southeast on the Albuquerque Basin of the Rio Grande rift, where wells draw from the Santa Fe Group aquifer system (Ritchie et al., 2019). The Jemez Mountains and the Sierra Nacimiento occupy the center of the county, and the northwest, including Cuba, lies on the eastern edge of the San Juan Basin, where water-bearing sandstones such as the Ojo Alamo Sandstone and the San Jose Formation dip into the basin (Renick, 1931; Kernodle, 1996; Thorn et al., 1990). Because these settings differ, this study compares wells by area and by named formation rather than treating the county as one aquifer.

2.2 Cuba Focus Area

Cuba lies on the eastern edge of the San Juan Basin, on the upper Rio Puerco; the river's USGS gage is 2.3 km from the village center. The case study focuses on the two aquifers mapped near the village (Sweetkind and Masbruch, 2019; Kernodle, 1996). On the USGS map of the Ojo Alamo Sandstone, Cuba sits at the southeastern end of the formation's outcrop, where the sandstone is at or near the surface (Thorn et al., 1990). North of Cuba, near Lindrith, the San Jose Formation contains four water-bearing units, and wells there produce water from its Cuba Mesa Member (Blandford et al., 2025). In 2020 the village pumped from an aquifer about 700 ft deep, with brackish water at greater depth (Davis, 2020); which formation its production wells draw from is not recorded in the public datasets. In this report, the *Cuba area* is the area within 25 km of the village center, and distances to Cuba are straight-line distances from that center.

3 Data

Town-specific data for Cuba are limited, so the study relies on public datasets that can be filtered to Sandoval County and the area around Cuba. Eight datasets were

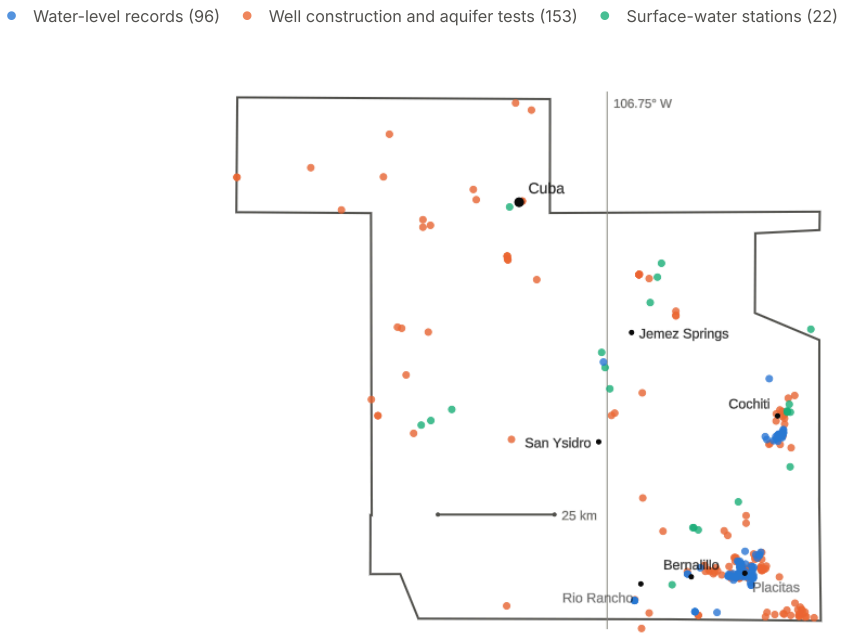


Figure 1. Every mapped site in the project's datasets, colored by what it measures: water-level records (USGS monitoring wells and NMBGMR wells), well construction and aquifer tests (OSE aquifer tests and well logs), and surface-water stations (USGS stream, canal, and reservoir sites, including the Rio Puerco at Cuba). Most sites lie east of 106.75° W along the Rio Grande; within 25 km of Cuba there are only one-time aquifer tests, one well log, and one stream gage. A dot is a site with any record, not necessarily active monitoring. Outline: Sandoval County (simplified, US Census Bureau). Town locations approximate.

collected (Table 1). For seven of them, the team wrote a one-page brief that records what the dataset contains, which research questions it can help answer, which plots to make from it, and what it can't show (Gonzales, 2026a,b,c,d,e,f,g); a plotting guide compares all seven (Gonzales, 2026h). The briefs and guide are published on the project's documentation site, and the raw files are kept unedited in the project repository (Djordjevic et al., 2026).

3.1 Datasets Used

Groundwater levels. Two datasets measure depth to water below land surface, where a larger number means a lower water table. The USGS statistics (U.S. Geological Survey, 2026c) give long monthly records for 9 wells at four sites: Sandia, Lincoln Middle School in Rio Rancho, Bernalillo, and Dome Road. The same download also covers stream gages, canals, and reservoirs, but only the groundwater records are used. The NM Bureau of Geology measurements (New Mexico Bureau of Geology and Mineral Resources, 2026c) cover more wells but fewer years: most were measured in 1997–1998, and only 5 wells were measured again in 2016–2019, so few of them can show a long-term trend.

Well and aquifer records. The State Engineer's aquifer-test records (New Mexico Office of the State Engineer, 2026b) are the only dataset with hydraulic properties, such as transmissivity and storativity, and they name the formation for 125 of the 179 tests. Seven test events name the Ojo Alamo Sandstone, among them 1957 and 1960 tests at what may be a single school well 0.7 km from Cuba, but only one names the San Jose Formation. The State Engineer's well logs (New Mexico Office of the State Engineer, 2026c) add borehole depth and the depth at which water was first found on the day each well was drilled, which is not a current water level.

Geology of the Cuba-area aquifers. Two publications describe the formations studied for RQ4: a USGS map atlas of the Ojo Alamo Sandstone across the San Juan Basin (Thorn et al., 1990), and a study of the San Jose Formation near Lindrith that identifies four water-bearing units and maps the Cuba Mesa Member (Blandford et al., 2025). Both are maps and reports rather than data tables, and neither shows which formation Cuba's municipal wells draw from.

Surface water at Cuba. USGS station 08332525 on the Rio Puerco (U.S. Geological Survey, 2026b) is the only dataset measured at Cuba itself. Six of its 10 streamflow visits found no flow, and the other four measured 6.9 to 20.4 cubic feet per second. Because these are occasional visits rather than a continuous record, the data are used

Table 1. Datasets collected for the study, with the team's brief for each where one was written. Distances are measured from Cuba.

Dataset	Source	Contents	Relevance to Cuba	Used for
Groundwater-level statistics (Gonzales, 2026a)	USGS	Monthly and yearly depth to water at 9 wells on 4 sites, 1998–2026	Wells are 65–95 km away, in the eastern part of the county	RQ1, RQ5
Groundwater levels (Gonzales, 2026b)	NM Bureau of Geology	456 depth-to-water measurements at 87 wells, 1997–2019, with well depth	Nearest well is 38 km away	RQ1, RQ2, RQ5
Aquifer tests	NM Office of the State Engineer	179 tests in Sandoval County, 1930–2008: well construction, formation, and hydraulic properties	Includes tests at a well 0.7 km from Cuba	RQ2–RQ4
Wells with well logs (Gonzales, 2026g)	NM Office of the State Engineer	Drilling records for 6 Sandoval County wells, 1978–2022: borehole depth and depth to first water	Link to Cuba's wells not verified	RQ2, RQ5
Ojo Alamo Sandstone (Gonzales, 2026e)	USGS map atlas, 1990	Maps of depth to the top of the sandstone, its thickness, and wells completed in it	Regional; covers the Cuba area	RQ4
San Jose Formation near Lindrith (Gonzales, 2026f)	NM Geological Society, 2025	Water-bearing units, thickness of the Cuba Mesa Member, and groundwater levels	Centered near Lindrith, about 30 km north	RQ4
Rio Puerco at Cuba (Gonzales, 2026d)	USGS station 08332525	10 streamflow measurements (2019–2026) and 4 gage-height readings	At Cuba, but measures the river, not groundwater	RQ3, RQ5
Aquifer sensitivity (Gonzales, 2026c)	NM Environment Department	Sensitivity class for 613 mapped areas	Can't be located: the file has no coordinates	Not used

only as context for the Cuba case study.

Not used. The aquifer sensitivity file (New Mexico Environment Department, 2026) has no coordinates or shapes, so it can't be mapped or matched to Cuba without its separate map layer, which was not obtained.

3.2 Data Not Available

The most important missing data are Cuba's municipal well records: well IDs, the formation each well draws from, pumping volumes, and water-level history, held by the Village of Cuba or the State Engineer (New Mexico Office of the State Engineer, 2026a). Six of the seven dataset briefs name these records as the missing piece for any conclusion about Cuba's own supply (Gonzales, 2026a,b,d,e,f,g). Also not obtained were the USGS digital aquifer data for the San Juan Basin (Sweetkind and Masbruch, 2019), groundwater-quality records such as the Brackish Water Assessment (New Mexico Bureau of Geology and Mineral Resources, 2026b), acequia locations (New Mexico Interstate Stream Commission, 2026), and records from other monitoring networks (U.S. Geological Survey, 2026a; New Mexico Bureau of Geology and Mineral Resources, 2026a). Section 7 describes what their absence means for the findings.

4 Methods

4.1 Data Cleaning

Cleaning makes the data correct and consistent without drawing conclusions; averaging, trends, and narrowing to the Cuba area happen in the analysis. Raw downloads are kept unedited. A MATLAB script, `cleaning/clean_csvs.m`, reads them and writes the cleaned files (Table 2), so anyone can rerun it and get the same result. As it runs, it prints checks such as row counts, duplicate rows, and the list of formation names.

Table 2. Cleaned files produced so far.

File	Rows	Contents
groundwater_levels_monthly.csv	2,673	Monthly mean depth to water at 9 wells, June 1998 to September 2026
aquifer_tests.csv	179	One row per aquifer test in Sandoval County, 1930 to 2008, with 17 columns

Shared conventions. Every cleaned file starts with the same six columns: site ID, site name, latitude, longitude, date, and source. Each has one row per measurement, sorted by site and date. Numbers are stored without units or commas, dates are written as YYYY-MM-DD, and column names end with their unit, such as `well_depth_ft`. Missing values are left blank rather than filled with placeholders such as 0 or −9999, which a chart would plot as real values.

Groundwater levels. From the USGS statistics, the script keeps the depth-to-groundwater rows that are monthly means and renames the columns to the shared names. Wells with an A or B suffix are kept as separate sites, because they share a location but are measured at different depths. The cleaned file was checked against the raw data: it has the same rows, dates, and values, with no duplicate site-and-date rows.

Aquifer tests. The script keeps all 179 Sandoval County tests, including repeat tests on the same well, and the 16 raw columns the analysis needs: well construction, formation, water level, pumping rate, and hydraulic properties. Empty and administrative columns are dropped. It converts numbers stored as text, standardizes dates, and cleans formation names by trimming spaces, converting them to uppercase, and fixing a misspelling. Three decisions were needed:

- One hydraulic conductivity was recorded as a range, 11 to 21 ft/day; the midpoint, 16 ft/day, is used.
- 104 tests list January 1, 1900, as the date of the starting water level, meaning the date is unknown, so that column is left out.
- 30 tests are dated January 1, which may mean only the year is known; they are kept as is.

Many tests are missing values, such as the formation (54 tests) and storativity (165 tests); these cells are left blank.

Open issues. Most of the State Engineer's well IDs in the aquifer-test file are incomplete: 157 of the 179 are partial, so the cleaned file has 73 site IDs for about 149 distinct wells. The analysis therefore tells wells apart by name and location rather than by ID. The Bureau of Geology water levels, the Rio Puerco measurements, and the well logs still need cleaning; the first and last of these also need their UTM coordinates converted to latitude and longitude. *[Update this paragraph once the remaining files are cleaned.]*

4.2 Analysis

The analysis focuses on measurable indicators, such as depth to water, water-level change over time, well depth, aquifer unit, and hydraulic properties from aquifer tests, rather than a full hydrologic model. Where records are sparse, the results say so instead of filling the gaps. All analysis code is in the `analysis/` folder of the repository. Each chart is a Python function that returns the figure together with its takeaway, caption, scope, and labeled findings, so the app, this report, and the documentation show the same numbers; every chart is reproduced on the documentation site's Analysis page. Datasets not yet cleaned (NMBGMR water levels, the Rio Puerco visits, and the OSE well logs) receive only minimal preparation in code, such as converting UTM zone 13 coordinates to latitude and longitude; the raw files are never changed.

Water-level trends (RQ1). Monthly mean depths to water were averaged into annual means, using only years with at least 9 months of data. Each well's trend is Sen's slope (the median of all pairwise slopes), and its significance is the Mann–Kendall test; a well is called falling or rising only when $p < 0.05$ (Helsel et al., 2020). Trends are reported for the full record and for 2016–2025. Because annual levels are serially correlated, the p-values and intervals are optimistic.

Wells and aquifer units (RQ2, RQ4). Each well contributes one point: the median NMBGMR reading, the first OSE aquifer test, or the depth where water was first found in a well log. Wells were grouped by the nearest of seven towns. Aquifer-test properties were compared by recorded formation, using one median per well. The scanned USGS map of the Ojo Alamo Sandstone (Thorn et al., 1990) was read by eye, not digitized.

Cuba case study (RQ3). The "Cuba area" is defined as within 25 km of the village center. For every dataset, we measured the distance from Cuba to its nearest site and recorded what kind of record it holds.

Coverage and indicators (RQ5, RQ6). Every site in every dataset was mapped and counted by year to show where and when monitoring exists, and each dataset was rated against each research question from its brief. Indicators were kept only where they can be computed from the data, each with the place it applies to and a stated confidence.

4.3 Classifying Findings

Each finding is labeled as one of the following:

- **Measured:** a trend or pattern the data show directly.
- **Hypothesis:** a possible explanation the data are consistent with but don't prove.
- **Insufficient data:** a question the available records can't answer yet.

4.4 Interactive Dashboard

Results are presented in an interactive Streamlit dashboard built with pandas and Plotly. Code and documentation are maintained on GitHub (Djordjevic et al., 2026). The dashboard uses the same chart functions as this report. Besides a home page, it has five pages: a county map of every site; the groundwater-level records and trends (RQ1); well depth, depth to water, and aquifer properties by area and formation (RQ2, RQ4); the Cuba case study (RQ3, RQ4); and data sources with monitoring coverage and the groundwater-stress indicators (RQ5, RQ6). Each chart is shown with its takeaway, its scope (Cuba, the Cuba area, or Sandoval County), a caption, its labeled findings, and the data behind it, which visitors can download. Charts can be zoomed and hovered for exact values. *[Dashboard lead: update this paragraph if the app's pages change before Nov 10.]*

5 Results

Table 3 summarizes the main result for each research question, and the subsections below give the details. Each finding is labeled *Measured*, *Hypothesis*, or *Insufficient data*, as defined in Section 4.

5.1 Groundwater Levels (RQ1)

The nine USGS monitoring wells sit at four sites between Rio Rancho and Cochiti, 65–95 km from Cuba, and have monthly records starting in 1998 or 1999 (Fig. 2). Over their full records, five wells fell, three rose, and one

Table 3. Main results by research question.

Research question	Main result	Label
RQ1. Groundwater levels	5 of 9 monitoring wells fell over 1998–2026, by up to 0.52 ft/yr; 4 of 7 were still falling in 2016–2025; the Sandia site has risen about 13 ft since 2007	Measured; cause of the Sandia recovery: Hypothesis
RQ2. Wells and aquifer units	Depth to water rises with well depth ($\rho = 0.63$–0.64); area medians range from 22 to 375 ft; median transmissivity is 136 ft^2/day, and within some formations it varies more than between formation medians	Measured
RQ3. Groundwater near Cuba	A baseline from 15 wells tested in 1957–1979, with water 18 ft below land 0.7 km from the village in 1957; the nearest repeated water-level record is 38 km away	Baseline: Measured; trend: Insufficient data
RQ4. Ojo Alamo vs. San Jose	The Ojo Alamo is shallow at Cuba and deepens northward; the one San Jose test (100 ft^2/day) matches the published Cuba Mesa Member value (101 ft^2/day)	Measured; statistical comparison: Insufficient data
RQ5. Monitoring coverage	86% of 271 sites lie east of 106.75° W; yearly water-level monitoring exists only since 1999, at four eastern sites, and none in the Cuba area	Measured
RQ6. Stress indicators	Five indicators computed from the data, each with its scope and confidence (Table 4)	Measured

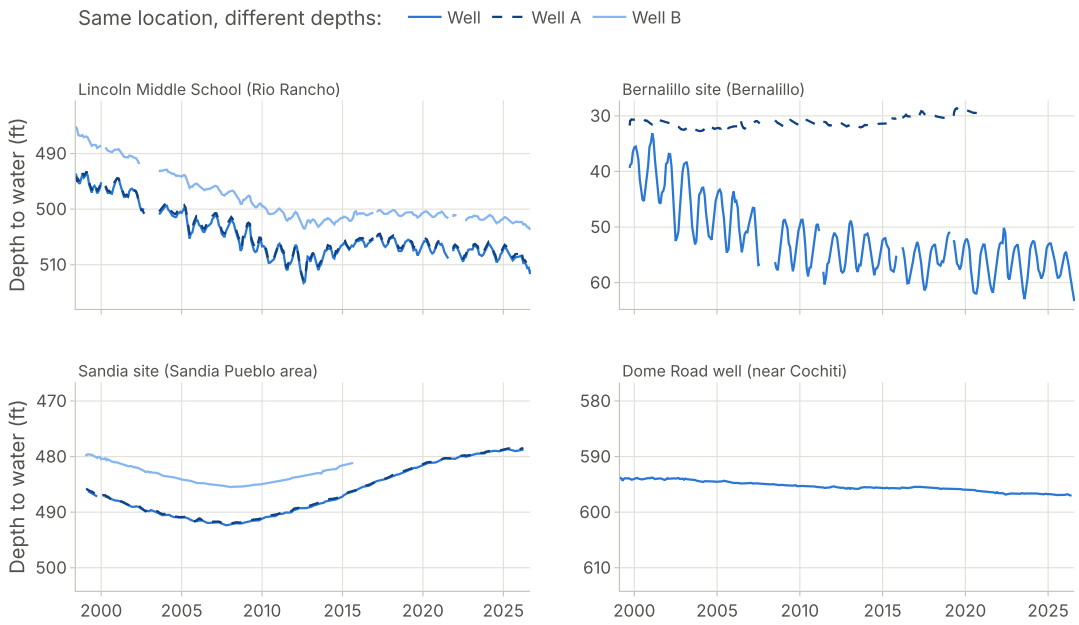


Figure 2. Monthly mean depth to water at the nine USGS monitoring wells, 1998–2026 (axis flipped: up is a higher water table). Wells marked A or B share a location but are measured at a different depth. Every panel spans the same vertical range, so steepness can be compared. Scope: Sandoval County, 65–95 km from Cuba. Source: USGS Water Data statistics, parameter 72019, cleaned by J. Wauneka.

showed no clear trend (Fig. 3). The three wells at Lincoln Middle School in Rio Rancho fell 0.36–0.40 ft per year and were still falling in 2016–2025, as was the Dome Road well near Cochiti. The Bernalillo site's main well fell 0.52 ft per year over its record, the steepest full-record decline, but showed no clear trend in 2016–2025. In all, four of the seven wells with enough recent data were still falling in 2016–2025, and the full-record change across the nine wells ranges from −16 to +8 ft (*Measured*).

The Sandia site behaves differently. Its water level fell until 2007 and then rose about 13 ft by 2025, at 0.72 ft per year in 2016–2025 (*Measured*). The recovery coincides with the start of the San Juan–Chama Drinking Water Project, after which the Albuquerque area's water utility pumped less groundwater and water levels rose across the Albuquerque area (Ritchie et al., 2019); these data alone cannot show the cause (*Hypothesis*).

Five NMBGMR wells in the Placitas area give an independent check over two decades. Measured in 1997–98 and again in 2018–19, three changed by 5 ft or less, one (PC-014) is 51 ft deeper, and one (PC-025) is 57 ft shallower (*Measured*). Two of the small changes, and PC-025's, start from 1990s readings flagged as recently pumped, which makes them less certain, and five wells describe those wells rather than an area-wide trend.

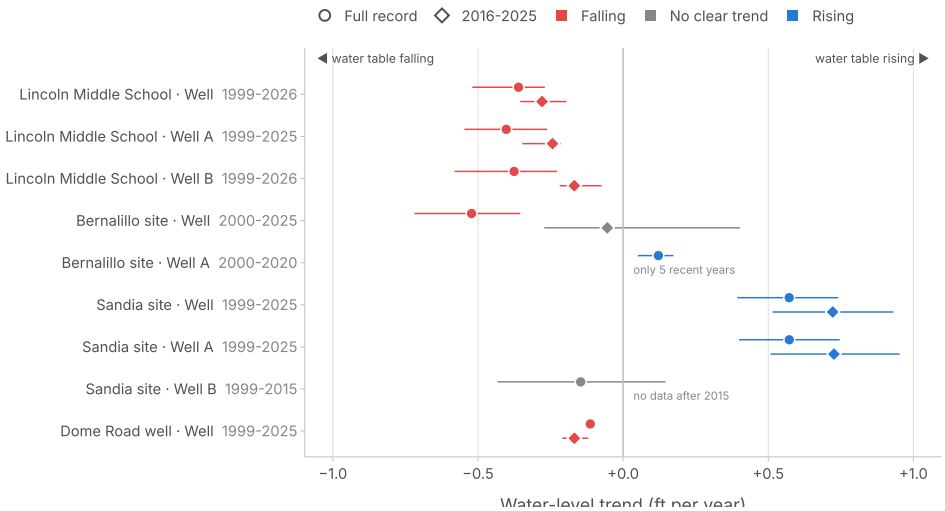


Figure 3. Water-level trend at each monitoring well (Sen's slope of annual means) over its whole record (circles) and over 2016–2025 (diamonds), with 95% intervals. Negative values mean a falling water table. Red: falling, blue: rising, gray: no clear trend (Mann–Kendall $p \geq 0.05$). Source: USGS Water Data statistics.

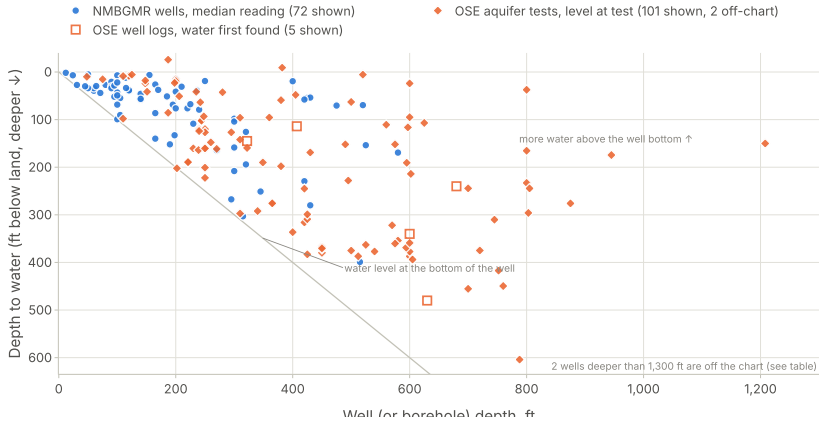


Figure 4. Well depth against depth to water, one point per well, from three record types, 1930–2022. The record types measure water differently (a 1997–2019 reading, the level on the day of an aquifer test, and the depth where water was first found during drilling), so they are context rather than one comparable series. Scope: Sandoval County, including 7 wells within 25 km of Cuba. Sources: NMBGMR groundwater levels; NM OSE aquifer tests and well logs.

5.2 Wells and Aquifer Units (RQ2)

Where water is deeper, wells are drilled deeper: the Spearman correlation between well depth and depth to water is 0.64 for NMBGMR wells and 0.63 for aquifer-test wells (Fig. 4). Half of the 180 wells with a recorded level have water 41–241 ft below land, and the median well has 142 ft of water standing above its bottom, a rough indication of how far the level could fall before the well would need deepening. Depth to water differs sharply by area: the median is 22 ft around Cochiti–Peña Blanca, 104 ft in the Cuba area, 116 ft around Placitas, and 375 ft around Bernalillo–Rio Rancho (*Measured*), although each area mixes record types and decades.

The aquifer tests add hydraulic properties (Fig. 5). Median transmissivity across 108 tested wells is 136 ft^2/day, with a middle half of 27.5–988 ft^2/day. Among formations with at least three tested wells, medians range from 6.2 ft^2/day in the Point Lookout Sandstone to 2,760 ft^2/day in alluvium and surface deposits, and within three formations the values vary more than the formation medians do, by almost five orders of magnitude in the Madera Limestone (*Measured*). The formation a well draws from matters, but so does the individual well. Only the aquifer-test file names the formation, so the other well records cannot be assigned to an aquifer unit.

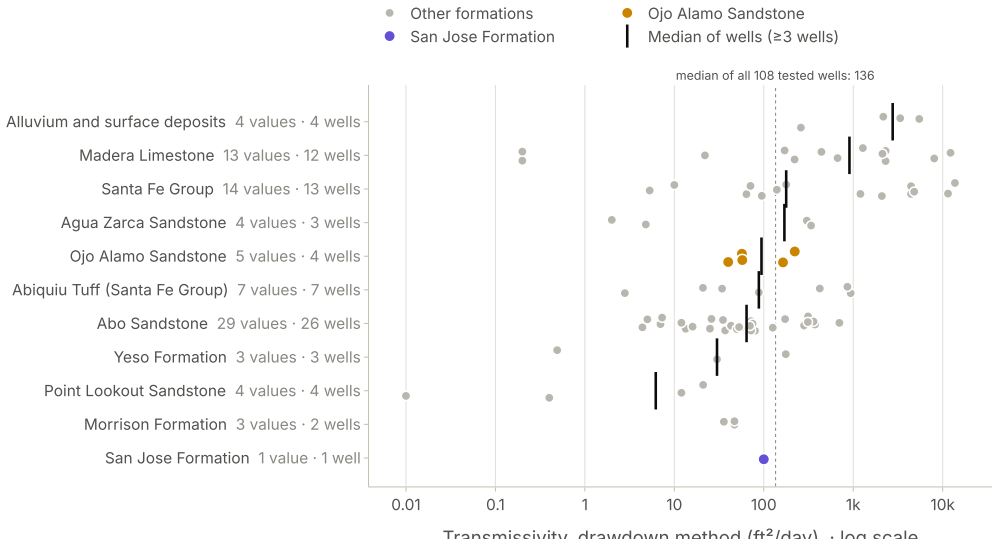


Figure 5. Transmissivity from each aquifer test (drawdown method), by geologic formation, on a log scale. Black ticks are formation medians (one median per well, formations with at least three tested wells); the dotted line is the median of all 108 tested wells, 136 ft^2/day. Tests date from 1930–2008. Source: NM OSE aquifer tests, cleaned by J. Wauneka.

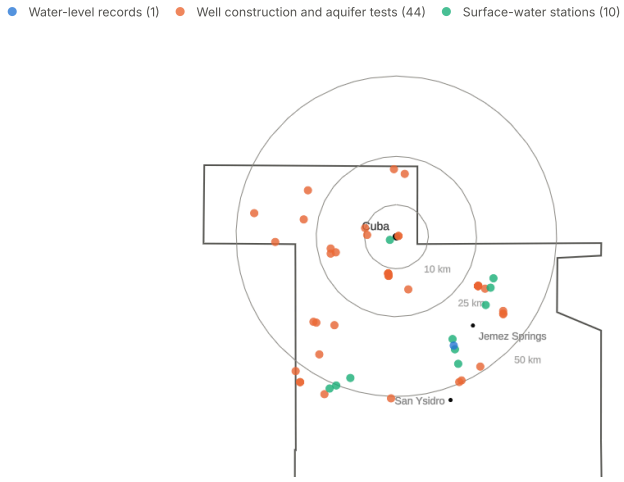


Figure 6. Every site within 60 km of Cuba in the collected data, colored by what it measures, with rings at 10, 25, and 50 km. Within 25 km, the records are one-time aquifer tests at 15 wells (1957–1979), one well log, and the Rio Puerco stream gage. Sources: USGS, NMBGMR, NM OSE; outline: Sandoval County (US Census Bureau).

5.3 Groundwater Near Cuba (RQ3)

Within 25 km of Cuba, the collected records hold 15 wells with aquifer tests from 1957–1979, one well log from 2018, and the Rio Puerco stream gage (Fig. 6). Nine of the tested wells recorded a depth to water on the day of the test (1957–1978), including 18 ft at a well 0.7 km from the village center in 1957. Across the ten Cuba-area wells with any recorded level (1957–2018), depth to water ranges from 10 to 509 ft, with a median of 104 ft (*Measured*). Together with the well depths, formations, and hydraulic properties from the same tests, these records form a historical baseline for the area.

None of these wells has been measured again in the collected data. The nearest well with repeated water-level measurements is 38 km away (two readings in 2017), and the nearest USGS monitoring well is 65 km away, so the public data cannot yet show whether Cuba's aquifer is declining, or how fast (*Insufficient data*). They do show what would answer the question: the village's own well records (well IDs, producing formations, pumping volumes, and water-level histories), and new water-level measurements at the baseline wells where they still exist, which would give the first direct estimate of long-term change near Cuba.

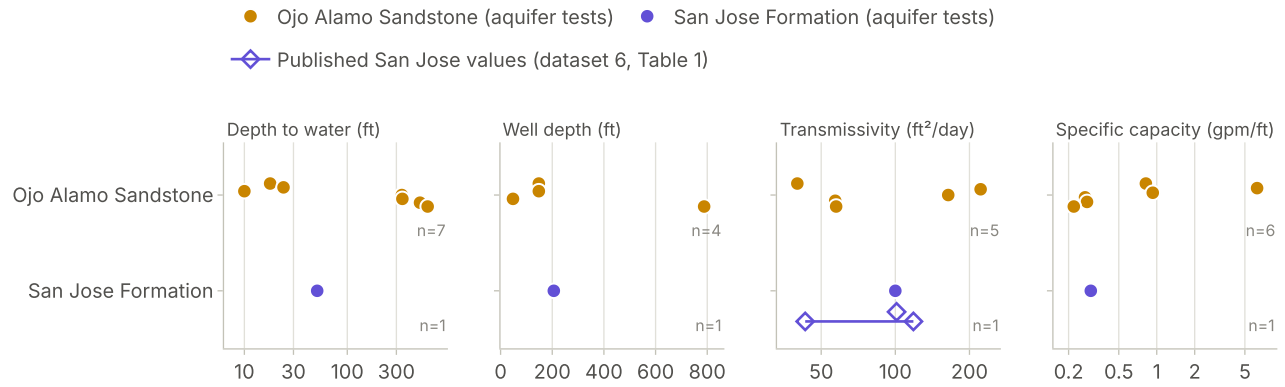


Figure 7. Every aquifer test that names the Ojo Alamo Sandstone (seven events) or the San Jose Formation (one), 1953–1977, with published San Jose transmissivity values compiled by Blandford et al. (2025) (open diamonds). Depth to water, transmissivity, and specific capacity use log scales. Scope: tests 1–45 km from Cuba. Source: NM OSE aquifer tests.

5.4 Ojo Alamo Sandstone and San Jose Formation (RQ4)

On the USGS map of the Ojo Alamo Sandstone (Thorn et al., 1990), Cuba sits at the southeastern end of the formation's outcrop, where the sandstone is at or near the surface, and its top deepens northward past 1,000 and 3,000 ft. In the aquifer tests, Ojo Alamo wells within 5 km of Cuba had water 18–24 ft below land in 1957–1960, while those 19–45 km away ranged from 10 to 604 ft (Fig. 7). The median transmissivity of the four Ojo Alamo wells with a value, 94.2 ft^2/day, lies within the middle half of all tested wells in the county (*Measured*).

The single San Jose Formation test, at a well 9 km from Cuba in 1977, recorded water 51 ft below land in a 206-ft well and a transmissivity of 100 ft^2/day. That value closely matches the one published value for the Cuba Mesa Member near Lindrith (101 ft^2/day) and falls within the range of earlier San Jose tests (43–118 ft^2/day), both compiled by Blandford et al. (2025): an independent check on the State Engineer's records (*Measured*). With seven Ojo Alamo test events and one San Jose test, all from 1953–1977, the two formations cannot be compared statistically (*Insufficient data*).

5.5 Surface Water at Cuba

The Rio Puerco at Cuba follows a clear seasonal pattern. It had measurable flow, 6.89–20.4 ft^3/s, on all four visits in April and May, and no measurable flow on any of the six visits from August to March, 2019–2026 (U.S. Geological Survey, 2026b) (*Measured*). The pattern fits a river fed mainly by spring snowmelt upstream (*Hypothesis*); ten spot visits cannot show how many days a year it runs dry.

5.6 Monitoring Coverage (RQ5)

The coverage analysis places every site from every dataset, 271 in all: 96 with water-level records, 153 with well construction or aquifer tests, and 22 surface-water stations (Fig. 1). Coverage is concentrated along the Rio Grande, where 86% of the sites lie east of 106.75° W. Groundwater levels have been measured every year only since 1999, and only at four sites in the east of the county; NMBGMR wells were measured in 1997–1999 and again in 2016–2019, with nothing in between (Fig. 8). The Cuba area has well records from nine separate years between 1957 and 2018 but no repeated water-level monitoring, and its only recent data are the Rio Puerco visits since 2019 (*Measured*). The dataset briefs reach the same conclusion: no dataset gives more than limited support for RQ3, the question about Cuba itself (Gonzales, 2026h).

5.7 Groundwater-Stress Indicators (RQ6)

For RQ6, we kept only indicators that can be computed directly from the collected data, and paired each with the place it applies to, its value, and a stated confidence (Table 4). The dashboard presents them the same way. None measures how much groundwater remains; together they show where groundwater stress can be measured and where it cannot yet.

6 Discussion

6.1 Persistent Declines, and a Recovery

Where the county is monitored, water levels are falling slowly but persistently. Rates of 0.36–0.52 ft per year at Rio Rancho and Bernalillo amount to roughly 4–5 ft per decade, and four of the seven wells with recent data were still falling in 2016–2025. These rates describe four sites rather than the whole county, but they are the only continuous multi-decade groundwater-level records in the collected data, and they come from the area where most of the county's people live.

The Sandia site shows that declines can reverse. Its

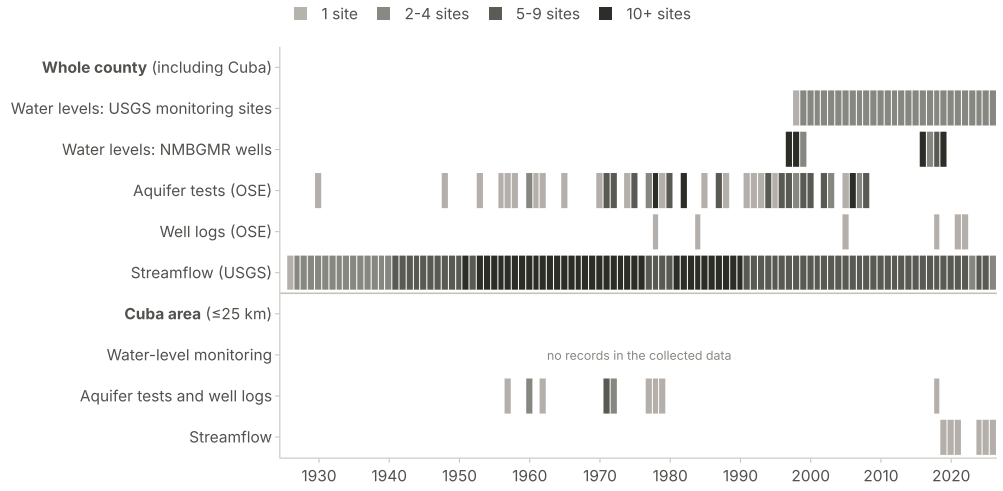


Figure 8. Number of sites with at least one record in each year, 1926–2026, for the whole county (top) and the Cuba area within 25 km (bottom). Darker shading means more sites. Gaps are years with no record in the collected files; other agencies may hold data. Sources: USGS, NMBGMR, and NM OSE.

Table 4. Groundwater-stress indicators that the collected data support, as shown in the dashboard.

Indicator	Applies to	Value	Confidence	Caveat
Water-level trend, 2016–2025	4 USGS sites, Rio Rancho to Cochiti	4 of 7 wells falling, 2 rising	Moderate (9–10 years per well)	Not representative of the whole county
Water-level change, full record	Same 9 wells	−16 to +8 ft, first to last full year	High at these wells	Depends on the start year (Sandia fell, then recovered)
Distance to nearest repeated water-level record	Cuba	38 km (2 readings, 2017)	High (a count, not an estimate)	Shows a data gap, not a water condition
Rio Puerco visits with measurable flow	Cuba (river)	4 of 10 visits (2019–2026)	Low	Spot visits; the season of each visit drives the result
Aquifer properties near Cuba	Cuba area (≤25 km)	15 tested wells, 1957–1979	Low	One-time historical tests; formation often not recorded

water level fell until 2007 and has since risen about 13 ft, beginning when the Albuquerque area's water utility started replacing part of its groundwater pumping with San Juan–Chama river water, and the USGS mapped widespread water-level rises across the Albuquerque area between 2008 and 2016 (Ritchie et al., 2019). Our data cannot prove the link, but the timing suggests that reducing pumping, through conservation or an alternative supply, lets water levels recover (*Hypothesis*). For communities facing decline, that is an encouraging precedent.

6.2 Local Conditions Differ

The well records also show why a county-wide average would mislead. Median depth to water differs by more than an order of magnitude between areas, and transmissivity varies by up to five orders of magnitude within a single formation. The condition of one community's aquifer cannot be inferred from another's, and well-specific records are worth collecting. This supports the study's two-level design: a county-wide screening to compare areas, and a local case study where decisions are made.

6.3 What the Results Mean for Cuba

For Cuba, the analysis turns an open question into a specific plan. The historical baseline shows that in the late 1950s water stood 18–24 ft below land in the Ojo Alamo Sandstone beside the village, and that the Ojo Alamo and San Jose formations in the Cuba region have transmissivities typical of the county's tested wells. The village's documented water problems from 2020 to 2026 involve infrastructure and water quality: damaged equipment, failed well pumps, deteriorated storage, and mineral-rich water (Office of the Governor of New Mexico, 2022; Hollinshead, 2021; U.S. Army Corps of Engineers, Albuquerque District, 2025; Village of Cuba, 2026a). Whether the aquifer itself is being depleted is a separate question that no public record yet answers.

Three steps would answer it at modest cost. First, compile the village's well records from the Village of Cuba or the State Engineer, including each well's producing formation, pumping volumes, and water-level history. Second, measure the static water level in the vil-

lage's wells and in any of the 1957–1979 test wells that can still be located, and compare it with the baseline. Third, add a well near Cuba to the state's groundwater monitoring network, which the state plans to expand with at least 100 new monitoring wells (Cook, 2026). Because the northwest of the county has no repeated water-level monitoring at all, one well there would likely add more information than another well along the Rio Grande.

6.4 Communicating Groundwater Stress

Of the indicators in Table 4, the water-level trend and the full-record change at monitored wells are the most direct measures of stress, with moderate to high confidence where they apply. The distance to the nearest repeated record is the clearest way to communicate a data gap: at Cuba it is 38 km. Showing each indicator with its scope and confidence, as the dashboard does, keeps a viewer from reading a measurement at Rio Rancho as a statement about Cuba (RQ6).

6.5 Peace Engineering

Disagreements over water often begin with disagreement about the facts. By combining scattered public records into one open, reproducible analysis, labeling every finding by how certain it is, and publishing the data, code, and dashboard, this project gives residents, the village, and state agencies a common starting point. It also makes an equity issue visible: monitoring is concentrated where most people live, while small rural communities such as Cuba, which rely on their own wells, have the least information about their aquifers.

7 Limitations

The results are bounded by the records available. The long-term water-level trends rest on four monitoring sites in the east of the county; they describe those sites, not a county average or the amount of water remaining. Most aquifer tests are decades old and record conditions on the day of the test, and only the aquifer-test file records the formation. No geologic-region layer was available, so wells are grouped by nearest town and by named formation. Trend tests assume that years are independent; serial correlation makes their p-values and intervals optimistic (Helsel et al., 2020).

The project proposal also asked about groundwater quality, acequias, and links between groundwater and surface water. These questions were set aside because the required data were not obtained: no groundwater-quality records were collected (the USGS water-quality measurements come from streams far from Cuba), no acequia data were available, and comparing groundwater with surface water requires repeated water-level measurements near a stream. Groundwater quality in particular, a documented concern in Cuba (Hollinshead, 2021), is a natural next step.

8 Conclusions

Using only public records, this study produced a reproducible, county-wide picture of groundwater conditions in Sandoval County and a historical groundwater baseline for Cuba. The main findings are:

1. Five of the nine long-term monitoring wells fell over 1998–2026, by up to 0.52 ft per year, and four of seven were still falling in 2016–2025.
2. At the Sandia site, the water table recovered about 13 ft after 2007, coinciding with a shift from groundwater to river water in the Albuquerque area.
3. Depth to water and aquifer properties vary strongly by area and within formations, so local decisions need local data.
4. Near Cuba, 15 wells tested in 1957–1979 provide a baseline, including shallow water in the Ojo Alamo Sandstone beside the village, and the one San Jose Formation test matches published values.
5. Repeated groundwater monitoring is concentrated along the Rio Grande; the nearest repeated record to Cuba is 38 km away.

We recommend that the Village of Cuba's well records be compiled and made public, that static water levels be measured in Cuba's wells and in the historical test wells, and that a monitoring well near Cuba be added to the state network. With those data, the workflow and dashboard built here can be rerun to answer whether Cuba's aquifer is declining.

Data and Code Availability

Code and data
github.com/pdjordj11/eng_220_aquifer_project
Interactive dashboard
eng220aquiferproject.streamlit.app
Documentation
pdjordj11.github.io/eng_220_aquifer_project
Dataset briefs and plotting guide
pdjordj11.github.io/eng_220_aquifer_project/data-collection

Author Contributions

P. Djordjevic: GitHub workflow, documentation, final report, presentation, and video.
D. Gonzales: data acquisition: collected seven datasets and wrote a research brief for each, plus a plotting guide comparing them.
J. Wauneka: data cleaning: wrote the MATLAB script that produces the cleaned groundwater-level and aquifer-test files.
M. Watkins: analysis code and visualizations: wrote the analysis package (data loaders, trend statistics, and 15 charts with labeled findings) that feeds the app, this report's figures, and the documentation site.
M. Yousef: Streamlit app design and development.

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